INCH-POUND

MIL-T-27/265B
16 November 2009
SUPERSEDING
MIL-T-27/265A
13 March 1991

MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP DOWN AND STEP UP

Inactive for new design
after 13 March 1991

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformer described herein shall consist of this specification sheet and MIL-PRF-27.

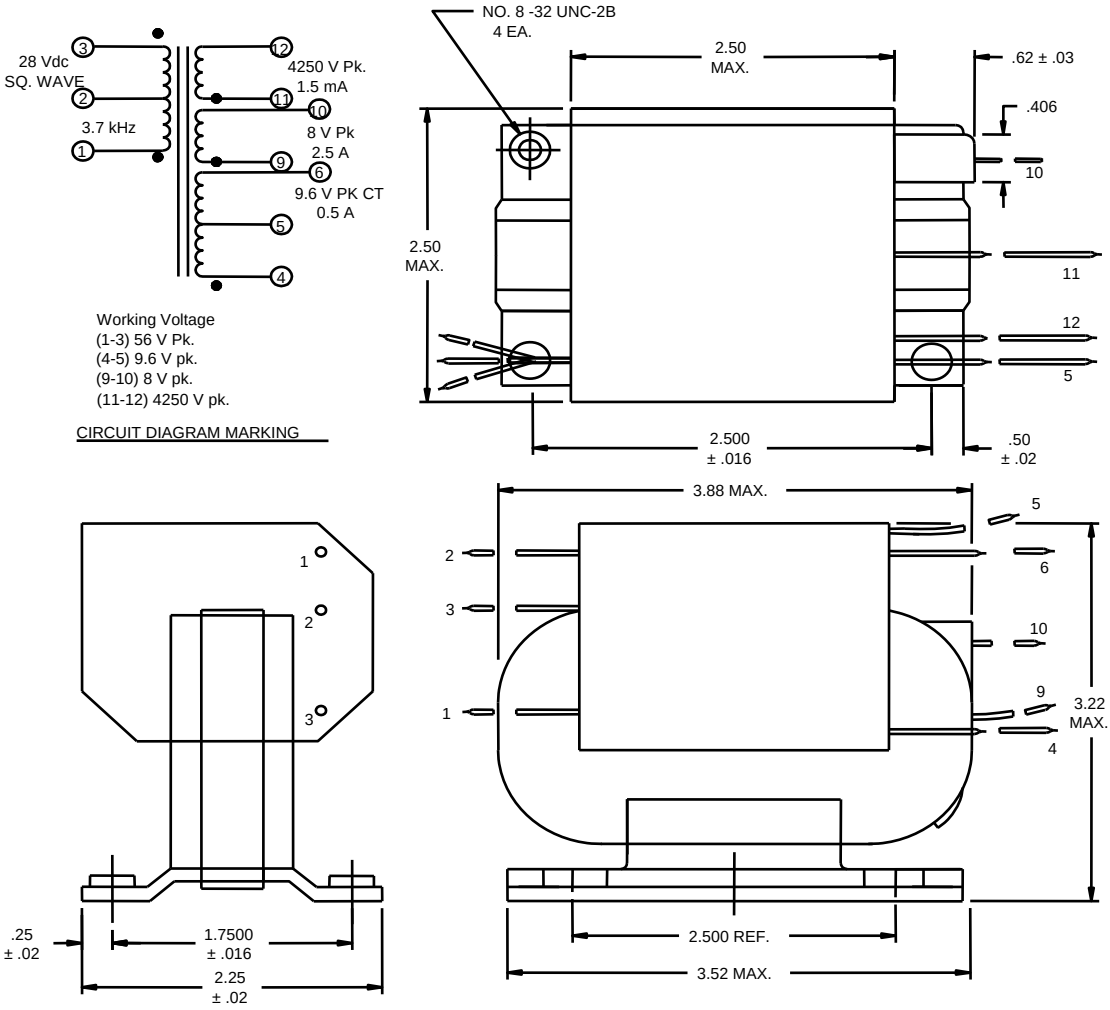

FIGURE 1. Dimensions and configurations.

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Inches	mm	Inches	mm
.016	0.41	2.25	57.2
.02	0.5	2.500	63.5
.03	0.8	2.60	66.0
.25	6.4	2.69	68.3
.406	10.31	3.22	81.8
.50	12.7	3.52	89.4
.62	15.7	3.88	98.6
1.750	44.45		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configurations – Continued.

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REQUIREMENTS: (When numbers in parentheses, such as (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: 3.7 kHz, square wave.

Primary (1-3): 56 V peak, 1.25 A, 35 volt-ampere.

Secondary :

(4-6): 9.6 V peak, 0.5 A rms.
(9-10): 8 V peak, 2.5 A.
(11-12): 4,250 V peak, 1.5 mA.

Working voltage:

(1-3): 56 V peak.
(4-6): 9.6 V peak, 0.5 A rms.
(9-10): 8 V peak, 2.5 A.
(11-12): 4,250 V peak, 1.5 mA.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Flex leads in accordance with SAE-AS81044/6

(1-3): AS81044/6-20-* or equal. 1/
((4-6), 11 and 12:): AS81044/6-18-* or equal. 1/
(9 and 10): AS81044/6-18-* or equal. 1/

Height:

(1): 10 inches, minimum.
(2): 9.5 inches, minimum.
(3): 11 inches, minimum.
(4-9): 6.50 inches, minimum.
(10): 7 inches, minimum.
(11): 4.5 inches, minimum.
(12): 5.5 inches, minimum.

Weight: 2.5 pounds, maximum.

Operating temperature range: -55°C to +130°C.

Altitude: 10,000 feet, maximum.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

1/ * Asterisks shall be replaced by color code designators in accordance with MIL-STD-681.

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Dielectric withstanding voltage (each winding):

At sea level:

11-9, 1, 4, and GND: 17.7 kV rms.
9-11: 13.4 kV rms.
1-4 and GND: 350 V.

Electrical characteristics: 2/

No load: With 28 volts rms, 3.7 kHz across (1-2), the current in (1-2) shall not exceed 400 mA rms and the power in (1-2) shall not exceed 11.2 watts.

Voltage across:

(1-3): .57 V ±5 percent.
(4-6): 0.102 V ±5 percent.
(9-10): .081 V rms ±2.5 percent.

Rated load: with 56 V peak, 3.7 kHz across (1-3), and the rated current on all secondaries.

Voltage across:

(4-5): 4.8 V peak ±5 percent.
(5-6): 4.8 V peak ±5 percent.
(9-10): 8 V peak ±5 percent.
(11-12): 8,500 V peak ±3 percent.

DC resistance:

(1-3): 0.163 to 0.22 ohms.
(4-6): 0.19 to 0.267 ohms.
(9-10): 0.016 to 0.0265 ohms.
(11-12): 2.2 k to 3.105k ohms.

Polarity: Additive with terminals 3 and 4 connected and terminals 3 and 11 connected (3 and 9 connected).

Temperature rise: 50°C with 56 V, 3.7 kHz across (1-3) at an ambient temperature of 80°C. Full load terminals are (4-6), (9-10), and (11-12).

Vibration (high frequency): MIL-STD-202, method 204, test condition A.

Shock (specified pulse): MIL-STD-202, method 213, test condition H.

VERIFICATION:

Qualification inspection: Not applicable.

Quality conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identification Number (PIN): M27/265-01.

2/ Apply 45 V, 400 Hz sinusoidal voltage to terminals 11 and 12 to verify the turns ratio. Turns ratio must not deviate more than 5 percent from calculated value.

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202
MIL-STD-681
SAE-AS81044/6

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2009-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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